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To Whom It May Concern,
Re: Reference on Diogo Poeta.

The present is a recommendation letter supporting Mr. Diogo Poeta Miguens.

I have known Diogo since the end of 2020, when he started his master project thesis "Extraordinary transparency in continuous metallic thin films".

His research was optimum and he showed a high interest on learn photonics and plasmonics phenomena.

In the project Diogo studied and design of a metasurface of bidimensional material transition metal dichalcogenide (TMDC), placed both on top and under the continuous metal film, obtaining a final geometry TMDC-metal-TMDC. Due to the anisotropic high refractive index of the TMDC metasurface, photonic-plasmonic resonances are created, which makes it possible to obtain high transmission through the continuous metallic film. By means, of finite-difference time-domain (FDTD) simulations, we engender configurations showing up to 90% transmission for a 30 nm thick metal film of silver, which is an enhancement ten times higher than the transmission of the metal without the nanostructure.

He learned techniques of 2D material characterization as Raman and photoluminescence spectroscopies together with other microscopies (SEM) and optical characterization techniques (ellipsometric). He also learned transfer techniques of bidimensional materials and he developed numerical systems based on the Lumerical FDTD software.

Diogo has shown initiative, independence, a solid research methodology and a high desire of working and learn.

Recently he has defended his project under my advice, which he has completed with honors.

On the whole, Diogo Poeta has demonstrated to acquire a solid basis in the subject matter of 2D materials and the physics that surrounds it, both in theory and in the lab, and in general he has demonstrated to have a strong capacity to quickly learn a new area. In his work he is a very interested and dedicated person and has proved to be a creative. From other points of view, he is a serious, sociable and agreeable individual to deal with. I am sure that he will very well adapt and perform in whatever new research group he starts a new undertaking. I recommend him.

Additionally, I would like to stress that Diogo Poeta is the kind of person that makes a group work better, as he (i) is very sociable and (ii) does not shy away from the communal work that arises in research groups.

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